EXECUTIVE SUMMARY

Paytm sees a shop's bad week before the loan payment fails. Sambhal helps instead of collecting.

RAVI'S MONDAY, AFTER 3 DAYS SHUT

Without Sambhal



₹3,400 · 57%

With Sambhal



₹1,032 · 17%

Share of the day's sales deducted. Without Sambhal: our model, as PhonePe documents.

HOW IT WORKS

Watch → **Talk** → **Fix** → **Verify**

Two days below half of normal sales: flagged before a miss

Voice note in his language, only 8 am–7 pm

Only the lender's pre-approved relief; code sets every rupee

Sales back, case closed, every step on record

22 → 5
merchants overdue, without vs with Sambhal (50 synthetic merchants, same days)

0 / 73
AI messages broke a rule (independent audit of the replay)

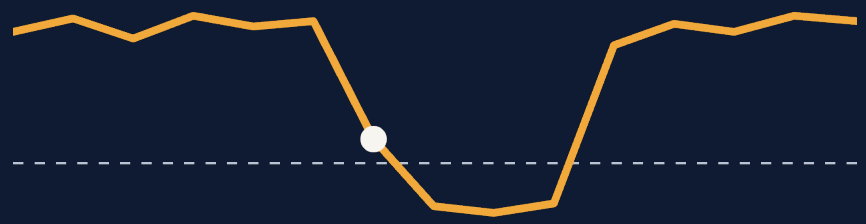
4 / 4
RBI no-contact occasions kept in code: bereavement, illness, family function, calamity

1 in 6,000
flagged cases kept from 90+ days overdue pays the AI cost of all 6,000*

WHY PAYTM ₹912 Cr a quarter in financial services revenue, +49%, led by merchant loans · first-loss guarantees up to ₹590 Cr

*About ₹10 of Sarvam calls a case (Ravi's 5 voice notes and 3 replies cost about ₹6). A loan like Ravi's (₹92,650 still owed) loses about ₹60,000 at 90+ days, where 30–35% is recovered. Sources: Paytm Q2 FY27 release (30–35%: p.15); Paytm filing, 26 Oct 2026, Annexure III (DLG). Demo figures are synthetic.

TRACK 2 · AI-POWERED FINANCIAL JOURNEYS



Sambhal

संभाल

Right-size the loan. Steady the merchant through bad weeks.

An AI teammate for Paytm merchant loans, before and during repayment

Siddh Mehta · Paytm Build for India AI Hackathon – Bengaluru · October 2026

THE PROBLEM

One bad week can sink a good loan

01

Sized on normal weeks

Installments follow normal sales. A wedding, an illness or a flood can shut the shop and drop settlements to near zero.

02

Shortfalls stack up

Unpaid installments carry forward (in our model, as PhonePe documents). On reopening day they are swept from sales or stay overdue on the record.

03

Outreach starts too late

Collections react to missed payments, not to the sales drop that predicted them. By then the talk is about arrears.

83–90%

of early missed-payment cases resolved (Bucket 1)

30–35%

recovered once a loan is 90+ days overdue

Timing decides the outcome. Help early, while it's still easy to fix.

Source: Paytm Q2 FY27 release p.15.

WHY NOW

The rules change on 1 January 2027

RBI's final recovery directions

RBI/2026-27/230 · issued 6 Aug 2026 · in force 1 Jan 2027

- Lenders must document engagement with distressed borrowers before escalating, and explain the resolution options
- Contact only between 8 am and 7 pm; calls recorded, and the borrower told
- No calls or visits during bereavement, medical emergencies, calamities or marriage functions
- Lenders pay borrowers for harm caused by rule-breaking recovery

Paytm is already in position

Paytm Q1 FY27 results · Pi launch, 2026

- Merchant loan distribution is one of Paytm's four growth engines
- Paytm acts as collection outsourcing partner on these loans
- Paytm already lists better collections as a goal for its AI
- Pi now offers AI agents to banks, lenders and insurers

Lending partners have ten weeks to build this. Sambhal's audit log is that record.

Sources: RBI/2026-27/230, Responsible Business Conduct amendment directions, 6 Aug 2026; Paytm Q1 FY27 earnings release; Paytm Pi launch coverage, Sep 2026.

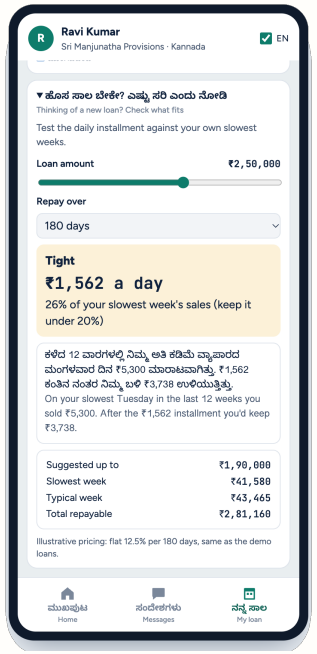
THE SOLUTION

One teammate, two risky moments

BEFORE THE LOAN

Right-size

- Tests the daily installment against the merchant's last 12 weeks of sales
- Shows what the installment leaves on the merchant's slowest days, in their language
- Suggests a size that keeps bad weeks manageable; the lender decides



DURING THE LOAN

Steady

1 · Detect

Flags two days below half of normal, before a payment is missed; spots whole-area dips

2 · Talk

Voice note or call in the merchant's language, inside the RBI rules

3 · Fix

Offers only lender-approved options: a short pause, smaller catch-ups, part-payment

4 · Verify

Watches sales recover, closes the case, then helps the merchant prepare

Sambhal never sets credit terms or invents offers. Lenders keep every decision.

HOW IT WORKS · DEMO STORY

A bad week, handled: Ravi’s kirana

Thu 9:05 pm

Detect

Second day below half (31%). Flagged before any miss; held till 8 am.

Fri 8:00 am

Talk

Kannada voice note, recording notice first: “Slow week? We can help.”

Fri 8:20 am

Understand

“Wedding at home, shop shut till Sunday.” No contact until Monday.

Fri 8:22 am

Fix

Lender allows a 3-day pause: ₹2,550 spread over 14 days. Ravi taps Accept.

Tue 8:00 am

Verify

Sales recover. Case closed, every step on record. Pay-ahead offered.

Monday’s deduction without Sambhal

₹3,400

57% of the day’s sales in one sweep (our model, as PhonePe documents)

→

With Sambhal

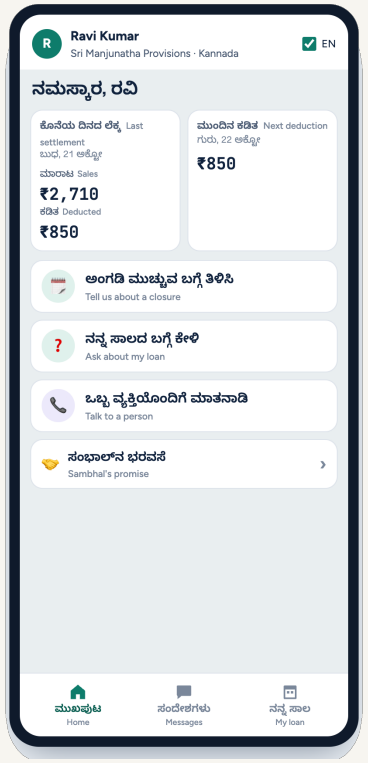
₹1,032

17%: the usual ₹850 plus ₹182 catch-up

Synthetic kirana, ₹850 a day, shut Fri–Sun. Without Sambhal the backlog is either swept from Monday’s sales or overdue on his record; Sambhal prevents both.

THE MERCHANT'S APP · WORKING PROTOTYPE

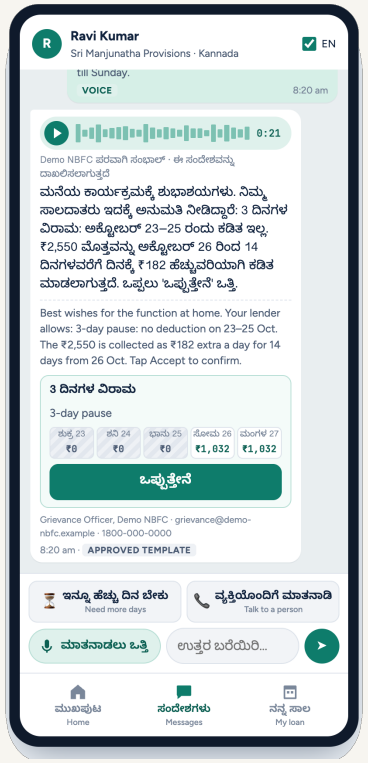
Built for a busy counter, in his language



Home: today's numbers and three big buttons



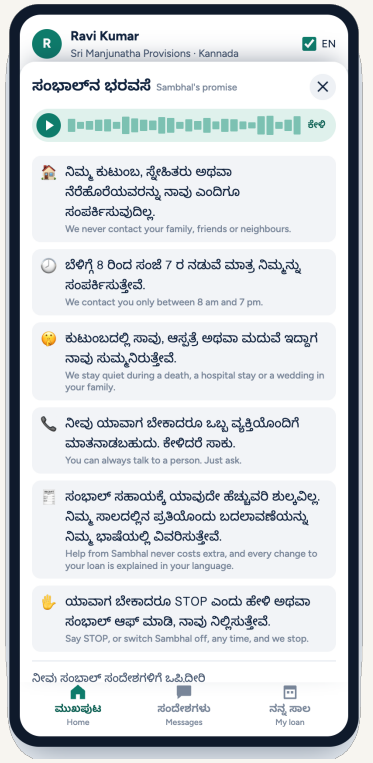
A voice note in Kannada; reply by voice or one tap



Only the lender's options, with a day-by-day preview



A calendar of paid, paused and catch-up days



Six promises, read aloud; switch off any time

PREPARE

Ready for the next bad week

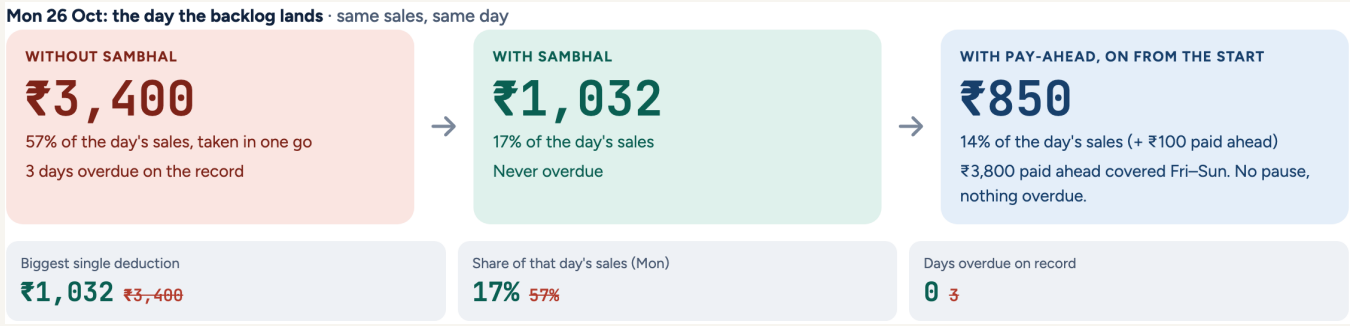
₹3,800

paid ahead by the wedding week, had Ravi switched on pay-ahead when his loan began

Good days: on days he sells ₹6,000 or more, ₹100 goes towards future installments (about 1.6% of a good day).

Bad days: the cushion pays first. Fri–Sun covered, nothing overdue, no lender decision needed.

His choice: off any time. Installments paid early to the lender, not a deposit or a wallet.



ACROSS THE SAME 50 MERCHANTS

If every flagged merchant had pay-ahead from the start: 1 merchant overdue for 1 day (22 without help, 5 with Sambhal's pauses).

An upper bound: it assumes everyone opts in. The pilot measures real uptake.

Sambhal doesn't just get a merchant through a bad week. It gets them ready for the next one.

GUARDRAILS

Built to say no

When this happens	Sambhal does this	Because
A message falls due at 7:30 pm	Holds it until 8 am the next day	RBI: contact only 8 am–7 pm
The merchant mentions a death or illness	Stops contacting, hands the case to a person for hardship help	RBI: no contact in bereavement or emergencies
The merchant asks to waive the dues	Declines kindly, offers only approved options, routes the request	The lender's policy, not the model, sets terms
Sales vanish with no reason given	Flags for human review and takes no action	Could be payments moved to another QR

Every message says it is recorded, and is logged. Before each release we replay 40 test situations; one rule break blocks it.

BEYOND ONE MERCHANT

What only Paytm can see

⚠ Neighbourhood alert · Bellandur

8 of 8 shops at 25% of normal on Wed 28 Oct, while 12% of shops elsewhere were down. Likely a local event (rain, road closure, power cut).

Sambhal has paused reminders there (RBI: no recovery contact during calamities) and asks the lender for one decision instead of separate check-ins.

Approve a 3-day pause for 8 shops

Manjunath Reddy 24% · Kavitha Naik 26% · Rakesh Yadav 26% · Ann Mathew 24% · Girish Bhat 26% · Salim Khan 25% · Priya Pinto 24% · Deepa Shetty 26%

Neighbourhood alert

Eight shops in Bellandur dip together while the rest of the city is normal. No reminders there (RBI: no contact in calamities); the lender approves one area pause in a click.

Tell us first

Shobha books her sister's wedding closure in the app before it happens. The pause is set for those days: no dip alarm, no check-in.

Trust, then check

Her shop reopens a day early. Sunday is collected as usual and her catch-up shrinks to the two days she was really shut.

Lender staff act inside the same rules: no calls at night or during a bereavement.

PROOF · SAME DATA, WITH AND WITHOUT

Same 50 merchants, same days

	Without Sambhal	With Sambhal
Merchants who fell overdue	22	5
Merchant-days overdue	66	32
Merchants 3+ days overdue	13	5
Biggest single deduction (median)	2.0×	1.2×
Rule breaks found by an independent audit	–	0 of 73 messages

40/40 scenario checks pass, live

160 automated tests

0 rule breaks found

Lender what-if: pause up to 2 days instead of 3, and 5 merchants overdue become 13. Lenders tune the menu with evidence before it goes live.

Synthetic merchants, 21 Oct–10 Nov. The 5 still overdue are with a person by design (bereavement, ICU, review, grievance, not eligible). This shows how a pilot would be measured, not a real-world result.

TECH STACK

How Sambhal is built

INPUTS

Daily settlements
Paytm merchant payments

Loan ledger & options
From the lending partner

Neighbourhood signal
Many shops in one area dipping together

THINK

Early-warning rule
Two days below half of normal, or anything overdue

Agent
Detect → talk → fix → verify → prepare

Language model
Sarvam-105b reads replies; keyword rules as a safety net

GUARD

Policy engine
Fixed rules the AI can't override

Contact rules
RBI hours, no-contact days, call limits

ACT AND PROVE

Indic voice
Sarvam Saaras v4 and Bulbul v3: Kannada, Hindi, English

Lender console
Cases, staff actions, alerts, audit trail

Eval harness
40 scenario checks, 160 tests; replay vs no Sambhal

Stack: Python · FastAPI · Sarvam AI (Saaras v4, Bulbul v3, Sarvam-105b) · plain JavaScript · pytest · runs offline

USP

Why this is hard to copy

1 A signal only Paytm has

Daily settlements from 1.64 crore merchants on subscription (Q2 FY27) show trouble days before a payment is missed. A lender without them sees only the miss.

2 Does what a risk score can’t

It works on top of the lender’s existing risk model: it holds the conversation, offers the fix and proves it worked.

3 Follows the rules by design

The 2026 RBI rules are written into the code, not left to the AI. Every message is timed and logged.

4 Proof, not promises

Results are compared with similar merchants who didn’t get Sambhal, and 40 test cases must pass before every release.

A score says who is at risk. Sambhal changes what happens next.

Merchant count: Paytm, Q2 FY27 (merchants on subscription).

IMPACT & BENEFITS

Better for merchants, lenders and Paytm

Merchant

- No backlog swept or left overdue on reopening day: 17% of the day's sales, not 57%, in our model
- Terms and options explained by voice, in their own language
- No calls at night, at a wedding or in grief

Lending partner

- Steps in at Bucket 1, where 83–90% of cases resolve
- Audit-ready for the 1 January 2027 rules
- AI conversations handle early cases before tele-calling starts

Paytm

- Better collection performance in its role as collection partner
- On-track merchants borrow again; repeat borrowers already get over half of disbursements
- A ready Pi module to sell to other lenders

PILOT SCORECARD Compared with merchants who don't get Sambhal: how many fall behind · cost per case fixed · complaints · rule breaks (target: zero)

Bucket-1 resolution: Paytm Q2 FY27 release p.15. Repeat borrowers: Paytm Q1 FY27 earnings release. Demo figures are synthetic.

BUSINESS MODEL

Lenders pay; merchants never do

Paytm's own lending partners

Bundled into Paytm's role as collection partner. Paytm earns through better resolution on loans it already distributes.

Banks and NBFCs, via Pi

A licence per active loan per month, plus an outcome fee for each account that gets back on track.

Compliance pack

The record RBI/2026-27/230 asks for: how the lender engaged a distressed borrower and which options it explained, before escalating.

PATH TO SCALE

1. Pilot: one city, one lending partner, a comparison group, eight weeks
2. Roll out across Paytm merchant loans
3. Offer through Pi to outside lenders, then extend to Postpaid and personal loans

Each case costs an AI conversation, not a field visit. We earn more only when merchants recover.

Paytm has approved up to ₹590 Cr of first-loss guarantees (DLG) for five lenders in FY27: loans that stay on track matter to Paytm directly.

DLG approvals: Paytm filing, 26 Oct 2026, Annexure III.

BUILT, TESTED, RUNNING

What you just saw is working

- ✓ 50 synthetic merchants, 5½ months of daily settlements: Ravi's week and 49 others
- ✓ Kannada, Hindi and English by voice with Sarvam, tested live; runs offline too
- ✓ A merchant app: voice notes, one-tap reasons, tell us first, pay-ahead
- ✓ A lender console: alerts, staff actions, what-if, audit log
- ✓ 40 scenario checks and 160 tests, run live; 0 rule breaks in the replay

TEAM

Siddh Mehta

ML Analyst, Airbus India, Bengaluru: reconciliation pipelines and NLP ticket clustering in production

Earlier: LangGraph agents at Fynd; document classification with Samsung PRISM

B.Tech CSE (Data Science), VIT Vellore

Sambhal: help before the backlog, and ready for the next one.

All merchant data in the prototype is synthetic. No real payments or messages are sent.

SARVAM AI · VOICE AND LANGUAGE

How we used Sarvam

Saaras v4 speech-to-text

The merchant answers by voice in Kannada, Hindi or English; the language is detected automatically.

Bulbul v3 text-to-speech

Every message is also a voice note in the merchant's language (voice: priya).

Sarvam-105b language model

Reads "wedding at home, shut till Sunday" as a reason. It never sets amounts or days: code does.

LISTEN-BACK: SARVAM CHECKS SAMBHAL'S OWN VOICE

Every saved voice note is played back through Saaras. A line whose amount or ending isn't heard as written is made again. Four faults found, four fixed:

WHAT WE HEARD		THE FIX
₹850 spoken as "800" in Kannada (every amount ending in fifty)	→	Amounts given to the voice in Kannada words, with **ರೂಪಾಯಿ**
Long Hindi lines cut off before the end	→	Spoken one sentence at a time
"सब ठीक है?" sounded like a statement	→	Hindi questions start with **क्या**
"23–25 Oct" heard as "23 and 25": Saturday sounded deducted	→	Spoken "from 23 to 25" in all three languages

Tested live at the venue: voice out 1.4 s · speech-to-text 0.8 s · language model 1.0 s. No key? The story's saved Sarvam audio still plays.